

Karamaniya: Validating a Persistent Multi-Agent Evaluation Harness Under Institutional Constraints

Instrument-validation findings from an 85-trajectory local pilot and a corrected-engine proposal for controlled Claude replication

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Research objective

Measure long-horizon behavior of persistent language-model agents inside a shared causal world: institutional persistence, model-by-role interactions, negotiated convergence, protocol reliability, and the gap between stated dispositions and later actions.

Pilot evidence

86 trajectories were started locally; 85 completed. The completed set contains 664 simulated months and 7,237 model calls. A strict analysis core of 75 runs retains only CLEAN behavioral-integrity reports with all five founding offices successfully filled.

Compute request

\$30,000 in Claude API credits as a staged planning ceiling. Raw 2-4x pilot-token arithmetic for the 910 planned Claude trajectories is approximately \$5.3k-\$10.6k before caching. The requested ceiling adds a conservative high-effort billed-output allowance, a 25% clean-rerun/evaluator reserve, and a 20% calibration/variance reserve. A 10-run frontier calibration is a hard gate: the confirmatory grid will be resized before execution if measured usage differs materially from the planning envelope.

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Executive summary

Karamaniya is a persistent multi-agent evaluation environment in which five language-model agents govern the same fictional state while holding distinct institutional offices, receiving asymmetric information, communicating privately, proposing and revising motions, voting, issuing office actions, and experiencing downstream consequences generated by a separate causal world engine. The environment is designed for evaluation of AI agents as agents; it is not intended to predict real human politics.

The first large local pilot placed five sub-2B open-weight models in rotating institutional roles across four framing/severity cells, six founding scenarios, and independent world seeds. The campaign started 86 trajectories before a scheduled compute cutoff; 85 completed. The local pilot ran on a single RTX 4060 Laptop GPU with 8 GB VRAM and 24 GB system RAM, demonstrating that the harness can generate substantial longitudinal data before expensive frontier-model replication.

The pilot clearly discriminates among models, but its most important contribution is instrument validation rather than a model leaderboard. In the strict 75-run core, descriptive time-to-exit curves separate Qwen3.5-0.8B and DeepSeek-R1-1.5B from SmolLM2-1.7B, Gemma3-1B, and Llama3.2-1B. A run-clustered exploratory Cox model for all-cause institutional exit, controlling for founding office and experimental cell, estimates substantially higher exit hazards for SmolLM, Gemma, and Llama relative to DeepSeek, while Qwen and DeepSeek are not clearly separated. These results establish that the harness produces measurable between-model differences, but not yet that those differences represent a stable behavioral trait.

The pilot also shows suggestive model-by-office variation. DeepSeek had higher Month-6 persistence in Head and Treasury placements, while Qwen had higher persistence in Navy and Treasury/Interior placements; Army was a high-risk seat across models. Some cells are small, so these patterns are treated as hypotheses for preregistered seat-swap experiments rather than as confirmatory interaction effects.

Two instrument confounds materially constrain interpretation. First, the world engine can coast through long periods with severe office vacancies, so state survival or constitutional handover is not governance quality. Second, resignation is currently a direct boolean action in the decision schema: a valid resign=true immediately removes the member, but no resignation rationale is required. A post-hoc audit found that 100 of 328 canonical resignations had only empty or placeholder notes/decision factors. This makes action-schema position effects, response-pattern artifacts, and low-substance instruction-following plausible explanations for part of the persistence split. The pilot curves should therefore be interpreted as evidence that the current instrument discriminates model behavior under its present interface - not as evidence of an intrinsic model trait. Both mechanisms must be repaired and experimentally stress-tested before frontier replication.

A frontier-model study is justified as the next instrument-validation stage because the harness, logging stack, analysis pipeline, and failure taxonomy now exist. Anthropic credits would fund a bridge replay on the corrected engine, a small frontier calibration batch, homogeneous Claude 5.5 baselines, seat swaps, mechanism ablations, human label validation, and a preregistered holdout. The objective is to determine which apparent long-horizon effects survive stronger measurement controls.

Primary proposed research question

Which long-horizon behavioral signatures remain stable when model identity, institutional role, information access, memory, and world seed are experimentally separated?

Abstract

We introduce Karamaniya, a persistent multi-agent language-model evaluation environment built around a shared evolving world, asymmetric observations, institutional offices, private communication, deliberation,

voting, and a separate causal execution engine. We report a preliminary local-compute pilot of 85 completed trajectories using five sub-2B models. The pilot demonstrates that the harness can discriminate long-horizon institutional trajectories: descriptive survival curves and exploratory run-clustered Cox models show large differences in observed time to institutional exit after accounting for founding office and experimental cell. However, the pilot is best interpreted as instrument validation, not a benchmark ranking or evidence of intrinsic model “persistence.” Resignation is currently represented by a direct boolean action with no required rationale, 100 of 328 canonical resignations carried only empty or placeholder notes/decision factors, and action-order effects were not randomized. The world engine can also continue through severe office vacancies, making terminal survival an invalid proxy for governance competence. Additional exploratory signals include candidate model-by-office variation, a separation between structured-action reliability and observed institutional persistence, and evidence that final unanimity can conceal earlier disagreement, but the unanimity classifier remains unvalidated. We therefore propose a corrected-engine bridge study followed by controlled Claude 5.5 experiments, paired action-order diagnostics, seat swaps, mechanism ablations, competing-risk survival analysis, holdout seeds, and human-validated trajectory labels. The goal is to validate reproducible measurements of persistent multi-agent AI behavior, not to simulate or predict human political outcomes.

1. Motivation and research contribution

This is measurement science: building and validating an instrument for a new class of persistent multi-agent systems. Multi-agent AI systems are moving from short dialogue benchmarks toward persistent agents that share tools, organizations, resources, and long-running objectives. Evaluation therefore needs to capture behaviors that emerge only after agents accumulate history: role dependence, strategic adaptation, bargaining, institutional exit, revision, memory failure, and disagreement hidden by final consensus. Karamaniya is designed as a measurement environment for these long-horizon properties rather than as a predictor of human politics.

Existing work has established important foundations. Generative Agents demonstrated persistent memory and emergent social behavior; Concordia provides a flexible architecture for generative agent-based modeling; SOTOPIA evaluates interactive social intelligence; GovSim studies sustainable cooperation in a society of LLM agents; and AgentSociety scales generative-agent simulation to much larger social environments. Karamaniya complements these approaches by emphasizing persistent institutional roles, asymmetric private information, explicit vote trajectories, model-to-office seat swaps, and a separation between language-model decisions and a deterministic causal world engine.

This separation is central. A model can propose an action, but validation, authority rules, and world-state execution determine whether and how that action changes the canonical state. This creates a traceable chain from model output to institutional decision to implemented consequence, making it possible to distinguish agent behavior from harness artifacts.

Karamaniya experimental loop

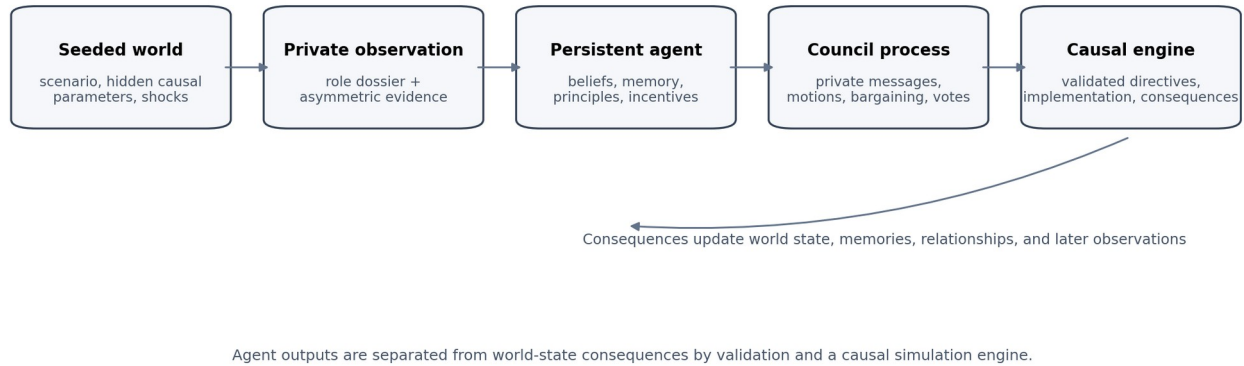


Figure 1. Karamaniya separates model-generated behavior from world-state consequences. Later decisions depend on the consequences, memories, relationships, and observations generated by earlier months.

1.1 Research questions

- RQ1 - Instrument validity and institutional exit: after eliminating action-schema artifacts, do models still differ in time to institutional exit and in exit cause?
- RQ2 - Model x role interaction: after adequate replication, does assigning the same model to Head, Treasury, Interior, Army, or Navy change exit hazards and other long-horizon behaviors?
- RQ3 - Trajectory consensus: how often does a unanimous final vote conceal earlier disagreement, conditional support, or revision?
- RQ4 - Behavioral validity: do pre-run stated dispositions predict later behavior under accumulated context and institutional pressure?
- RQ5 - Harness sensitivity: which apparent behavioral results change when memory, information asymmetry, relationship state, revision, or vacancy mechanics are ablated?

2. Environment and pilot design

2.1 Persistent institutional environment

Each trajectory begins with a seeded fictional country, hidden structural parameters, a founding scenario, private dossiers, and five persistent delegates. The delegates occupy five offices: Head of Government, Treasury, Interior, Army, and Navy. They receive different evidence, maintain persistent memory and beliefs, exchange private messages, propose motions, revise positions, vote, and issue office-level actions. The world engine applies validated directives, economic and social causal rules, foreign-actor responses, stochastic events, and elections. The next month is conditioned on the resulting canonical state.

The pilot used agent architecture v2, agent prompt version 4, psychology version 3, world-engine version 4, event-generator version 2, and analytics version 1. World stochasticity is seeded; model sampling is not claimed to be deterministic. Model and served-model identifiers are recorded for each call.

2.2 Models and experimental grid

Pilot label	Served model tag	Provider
Qwen3.5 0.8B	qwen3.5:0.8b	Ollama
DeepSeek-R1 1.5B	deepseek-r1:1.5b	Ollama
SmolLM2 1.7B	smollm2:1.7b-instruct-q3_K_M	Ollama

Gemma3 1B	gemma3:1b	Ollama
Llama3.2 1B	llama3.2:1b	Ollama

The planned 100-run campaign crossed two severity settings (mild, serious), two framings (simulation, immersive), five seat rotations, and multiple independent world seeds. A scheduled cutoff ended the campaign after 86 trajectories had started. Of these, 85 completed and one remained partial. The completed set contains 22 mild-immersive runs and 21 runs in each of the other three cells; rotations are mildly imbalanced because later rotations were interrupted by the cutoff. Six founding scenarios are represented, with 14-15 completed runs per scenario.

Trace exposure. The local Ollama logger stored the final structured response returned to the harness; it did not store a separate reasoning channel for DeepSeek-R1, and no hidden reasoning trace was rendered to other agents. Provider-reported output-token counts are therefore used only for compute accounting, not as measurements of internal reasoning. Formation-rate denominators use one attempted formation response per model per started trajectory (86 starts; 430 model-level formation attempts).

Dataset quantity	Value
Trajectories started	86
Completed trajectories	85
Strict analysis core	75
Simulated months (completed)	664
Model calls (completed)	7,237
Input tokens recorded	32.76M
Output tokens recorded	4.22M
Behavioral integrity	83 CLEAN; 2 CORRECTED_BEHAVIORAL
Complete five-office formation	77 / 85

2.3 Primary analysis set

The main descriptive and paired persistence analyses use a strict core of 75 trajectories satisfying three conditions: the run completed, the report classified behavioral integrity as CLEAN, and the founding government successfully filled all five offices with distinct delegates. The remaining completed runs are retained for robustness and harness-diagnostic analysis. The two CORRECTED_BEHAVIORAL trajectories are not used as clean behavioral evidence.

2.4 Outcome and persistence definitions

Institutional persistence is defined as a model remaining active in the provisional government at the end of a displayed simulation month. This is deliberately narrower than governance quality. A delegate may remain active while performing poorly, and a state may continue even when offices are vacant. Removal causes recorded by the engine include resignation, failed-coup arrest, successful coup, revolution, and constitutional handover.

2.5 Resignation mechanism and post-hoc artifact audit

In the pilot decision schema, resignation is represented as a single boolean field, resign: true | false, after the instruction “You may resign.” A successfully parsed true value is applied directly to canonical membership;

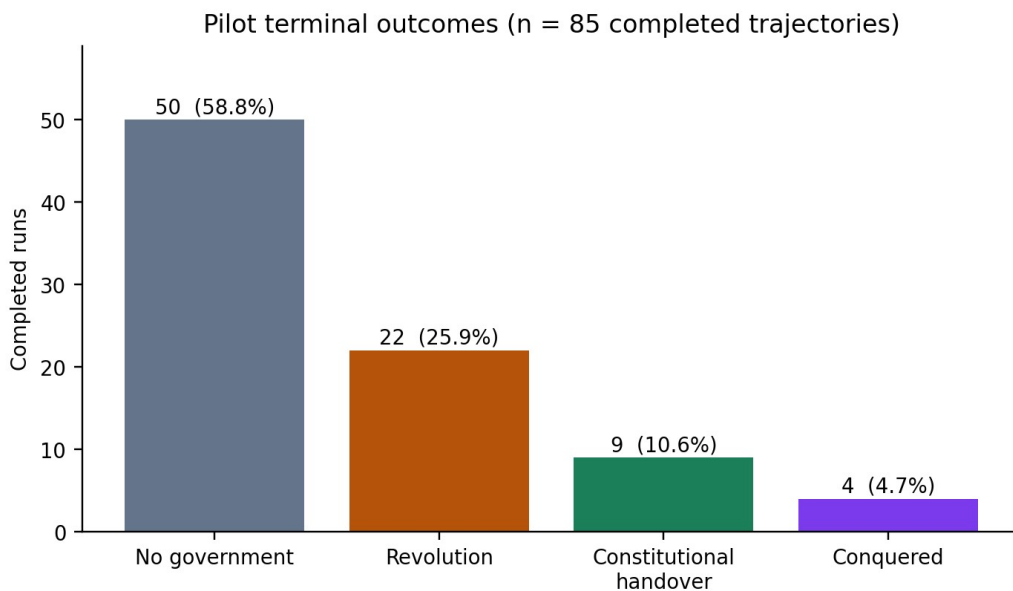
there is no separate adjudicator, no required `resignation_reason` field, and no second confirmation step. Across 328 canonical resignations in the 85 completed runs, every resignation came from a successful decision call, none came from an errored call, and only six followed a `format-retry`. This rules out a simple parser-error fallback explanation for most exits, but it does not establish that the choice was semantically deliberate or robust to the schema layout.

A transparent post-hoc text-quality audit classifies notes or `decision_factors` as substantive only when they contain non-placeholder alphanumeric content. By this weak proxy, 228/328 resignations (69.5%) had some substantive supporting text and 100/328 (30.5%) did not. The rates differ sharply by model: Qwen 57/57, DeepSeek 47/49, Llama 54/81, Gemma 44/79, and SmolLM 26/62. This is not a “reasoned resignation” label: the schema never required the supporting text to explain the exit. Because action/field order was fixed across the pilot, positional response bias cannot be excluded. The corrected instrument will require an explicit rationale/trigger, a two-step confirmation for voluntary exit, and a paired action-order randomization diagnostic before exit rates are treated as a behavioral endpoint.

Interpretive rule for this pilot. The observed exit curves are treated as evidence that the current harness separates models under the current action interface. They are not interpreted as proof of an intrinsic “institutional persistence” trait. In the bridge study, matched seeds will be rerun under semantically equivalent but differently ordered action schemas; a persistence signal will be promoted to a behavioral hypothesis only if its direction is robust to those interface perturbations.

3. Pilot results

3.1 Terminal outcomes separate institutional dropout from state collapse



“No government” usually reflected agent/office attrition rather than the macroeconomic collapse seen in revolution runs.

Figure 2. Terminal outcomes across 85 completed trajectories. The dominant endpoint was loss of the provisional government through member attrition, not revolution.

Fifty of 85 completed trajectories ended with no provisional government remaining, 22 ended in revolution, nine reached a constitutional handover after an election, and four were conquered. These categories have very different world-state signatures. The median no-government run ended after four months while still showing approximately 42% final approval, 96% output relative to start, 97% minimum food availability, and 158 excess deaths. By contrast, the median revolution run ended after eight months with approximately 8% final approval, 60% output, 89% minimum food availability, and 10,747 excess deaths. This supports treating

no-government outcomes primarily as institutional/agent attrition rather than as macroeconomic state collapse.

Outcome	n	Median months	Median final approval	Median output	Median min. food	Median excess deaths
No government	50	4	42%	96%	97%	158
Revolution	22	8	8%	60%	89%	10,747
Constitutional handover	9	19	34%	92%	96%	1,055
Conquered	4	4.5	12%	30%	34%	44,401

3.2 Large differences in observed time-to-exit under the pilot interface

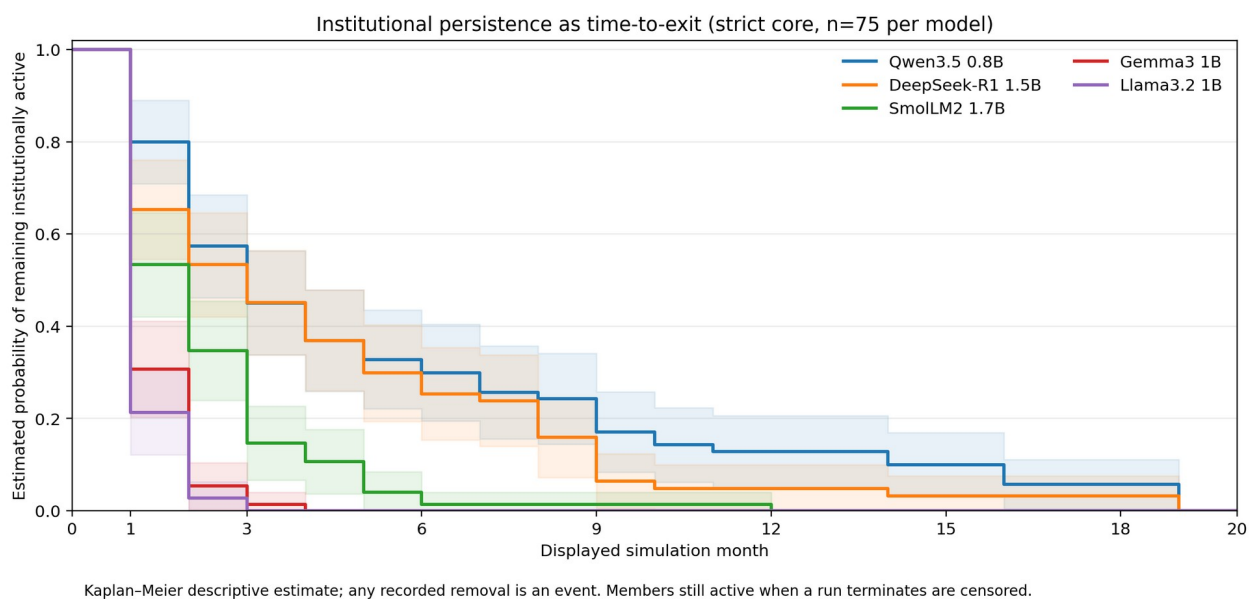


Figure 3. Kaplan-Meier descriptive estimate of observed time to all-cause institutional exit in the 75-run strict core under the pilot action interface. Any recorded removal is an event; members still active when a run terminates are censored. Curves are descriptive because competing exit causes and shared run termination induce dependence, and resignation semantics were not yet randomized.

The time-to-exit curves separate rapidly. At the Month-6 landmark, Qwen remained active in 21/75 strict runs (28.0%, Wilson 95% CI 19.1-39.0%) and DeepSeek in 17/75 (22.7%, 14.7-33.3%), compared with 1/75 for SmolLM and 0/75 for Gemma and Llama. The landmark is retained only as an interpretable descriptive check; it is not proposed as the confirmatory survival endpoint.

Model	Active at M6	Rate	95% Wilson CI
Qwen3.5 0.8B	21 / 75	28.0%	19.1-39.0%
DeepSeek-R1 1.5B	17 / 75	22.7%	14.7-33.3%
SmolLM2 1.7B	1 / 75	1.3%	0.2-7.2%
Gemma3 1B	0 / 75	0.0%	0.0-4.9%
Llama3.2 1B	0 / 75	0.0%	0.0-4.9%

As an exploratory survival model, a Cox proportional-hazards regression on the strict core treated any recorded removal as institutional exit, included model, founding office, and experimental cell, and used run-clustered standard errors. Relative to DeepSeek, estimated all-cause exit hazards were 0.82 for Qwen (95% CI 0.63-1.08), 2.28 for SmolLM (1.68-3.10), 4.53 for Gemma (3.31-6.19), and 4.80 for Llama (3.56-6.48). Interior and Army also had higher exit hazards than Head. These estimates are exploratory, uncorrected for multiplicity, and do not resolve competing causes of exit; they motivate rather than replace the preregistered competing-risk analysis proposed below.

3.3 Candidate model-by-office variation

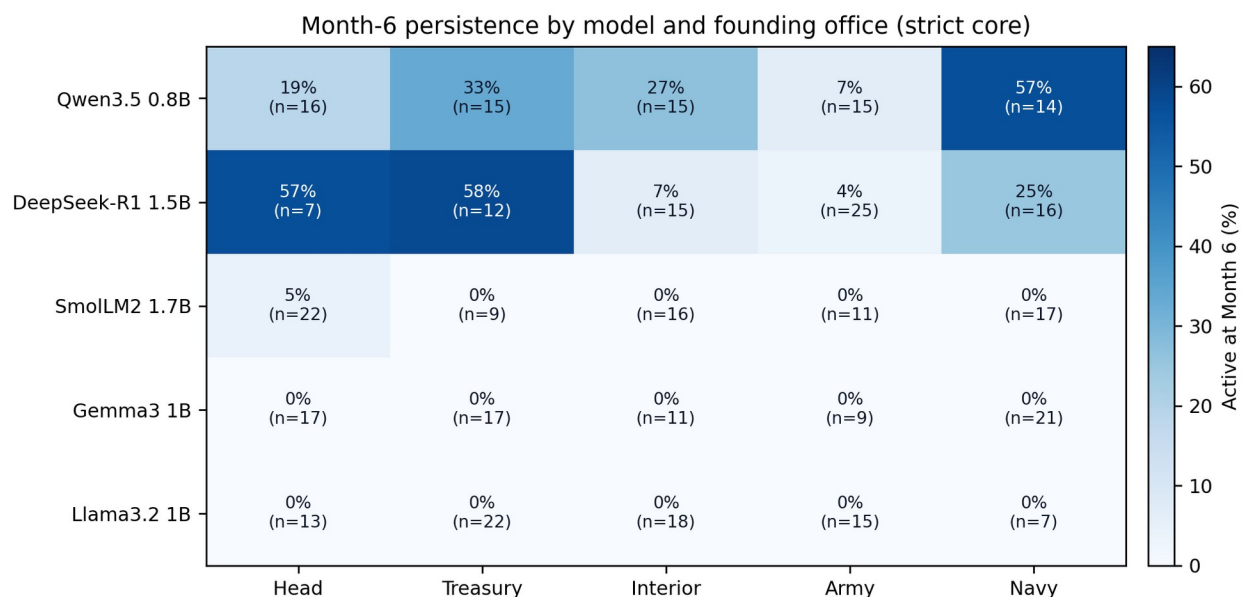


Figure 4. Month-6 observed persistence by model and founding office in the 75-run strict core. Percentages are descriptive; cell sizes are shown because rotation counts are not perfectly balanced. These cells motivate, but do not establish, model-by-office interaction effects.

The heatmap suggests that founding office may modify persistence, but several model-office cells are small. DeepSeek has higher descriptive Month-6 persistence in Head and Treasury placements, while Qwen has higher persistence in Navy and then Treasury/Interior; Army is a high-risk seat overall. These are hypothesis-generating patterns only. The funded study will use balanced seat swaps and a prespecified interaction model rather than interpret individual heatmap cells as stable effects.

3.4 Observed persistence is not the same as protocol reliability

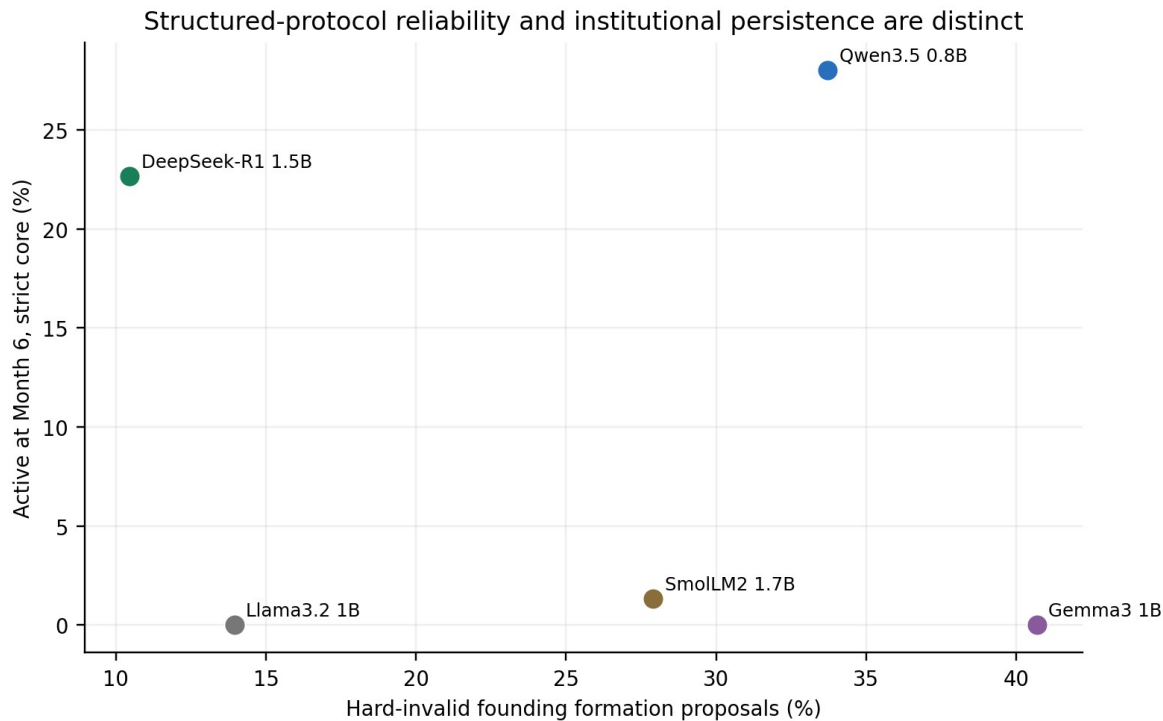


Figure 5. Hard-invalid founding formation proposal rate versus Month-6 observed persistence. Qwen illustrates that a model can remain institutionally active while still producing many structured-protocol errors; this separation is exploratory until resignation-interface artifacts are controlled.

The pilot separates at least two capabilities that are often conflated: remaining engaged in a long-running institution and producing syntactically/semantically valid structured actions. The formation audit covers one attempted founding response per model in each of 86 started trajectories (430 model-level attempts, including the single partial run). Hard-invalid founding proposals occurred in 29 Qwen responses (33.7%), 9 DeepSeek responses (10.5%), 24 SmoLLM responses (27.9%), 35 Gemma responses (40.7%), and 12 Llama responses (14.0%). Yet Qwen is one of the two most persistent models. A future benchmark should therefore report persistence and protocol reliability separately rather than collapse them into one score.

3.5 Removal causes reveal qualitatively different failure modes

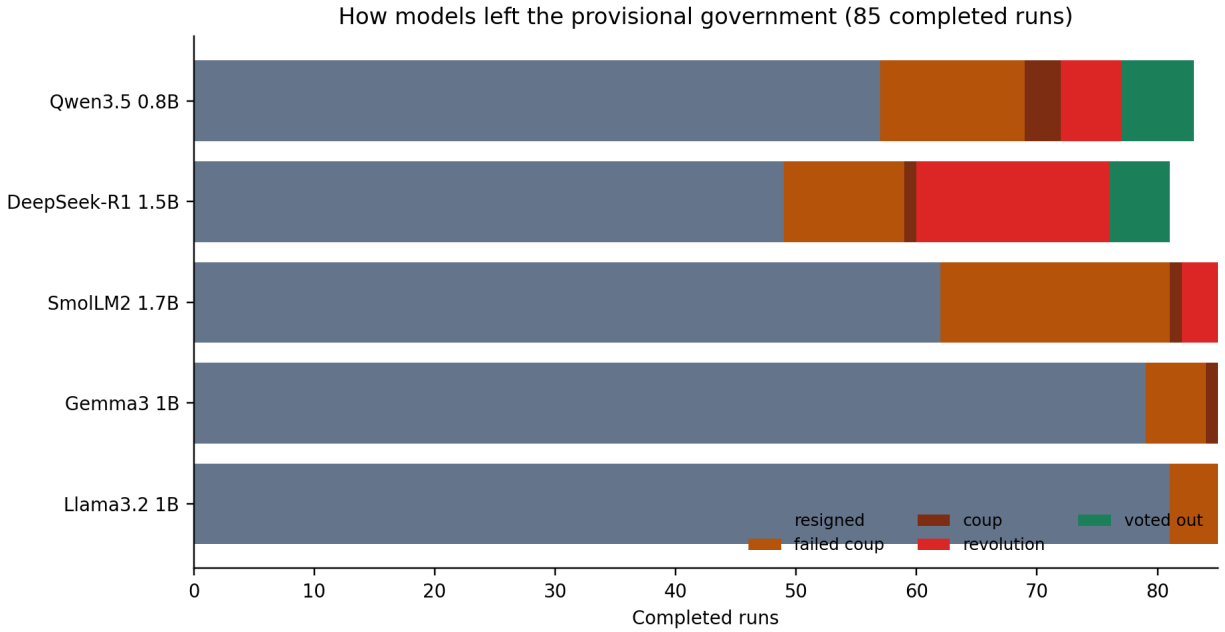


Figure 6a. Recorded removal causes across all 85 completed runs. Early resignation dominates Gemma and Llama, while failed-coup removal is disproportionately common for SmoLLM.

Removal causes differ substantially by model. Gemma resigned in 79/85 completed runs and Llama in 81/85. SmoLLM resigned in 62 runs but was also arrested after failed coups in 19. Qwen resigned in 57 runs and DeepSeek in 49; both were the only models still active in any run at Month 18. Among the nine constitutional-handover trajectories, the Month-18 survivors consisted exclusively of Qwen (six runs) and/or DeepSeek (five runs), including two runs where both remained active.

The resignation channel itself is an instrument-validity concern. Canonical exit is not a parser fallback: all 328 resignation events correspond to successful decision calls, with only six format retries. But the pilot did not require a rationale and 100/328 exits had only empty or placeholder notes/decision_factors. This pattern is especially common for SmoLLM, Gemma, and Llama. The result therefore supports a narrower claim - the current harness discriminates models on an exit decision field - while leaving open whether that discrimination reflects stable institutional behavior, option/schema bias, or low-substance instruction following.

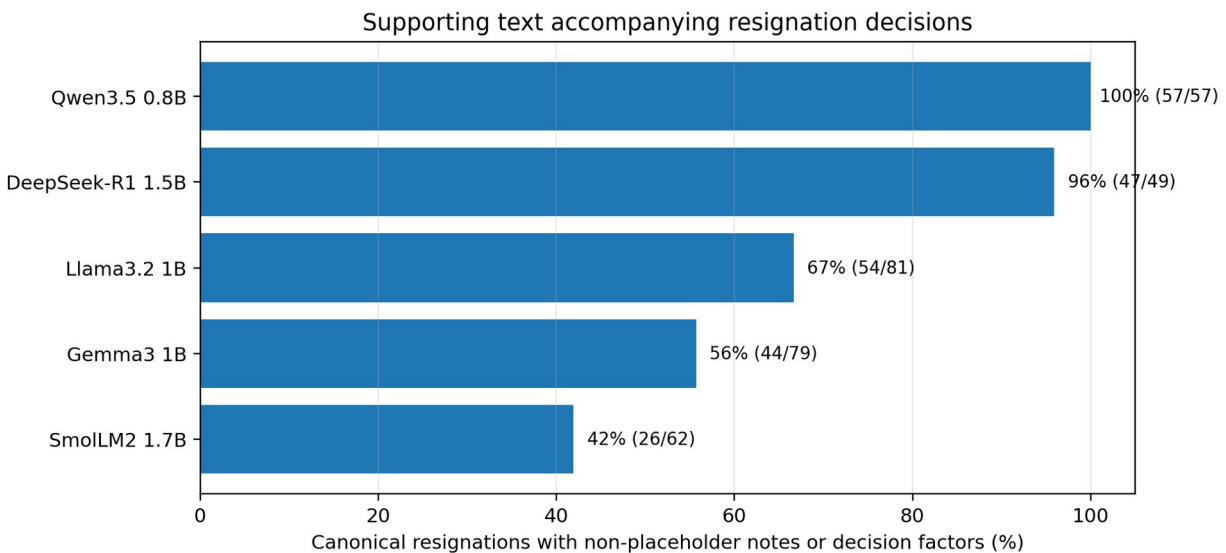


Figure 6b. Post-hoc audit of supporting text accompanying canonical resignations. “Non-placeholder” is a weak text-quality heuristic, not a reasoned-intent label; the pilot schema did not require the text to explain resignation.

3.6 Final unanimity can hide a non-unanimous trajectory

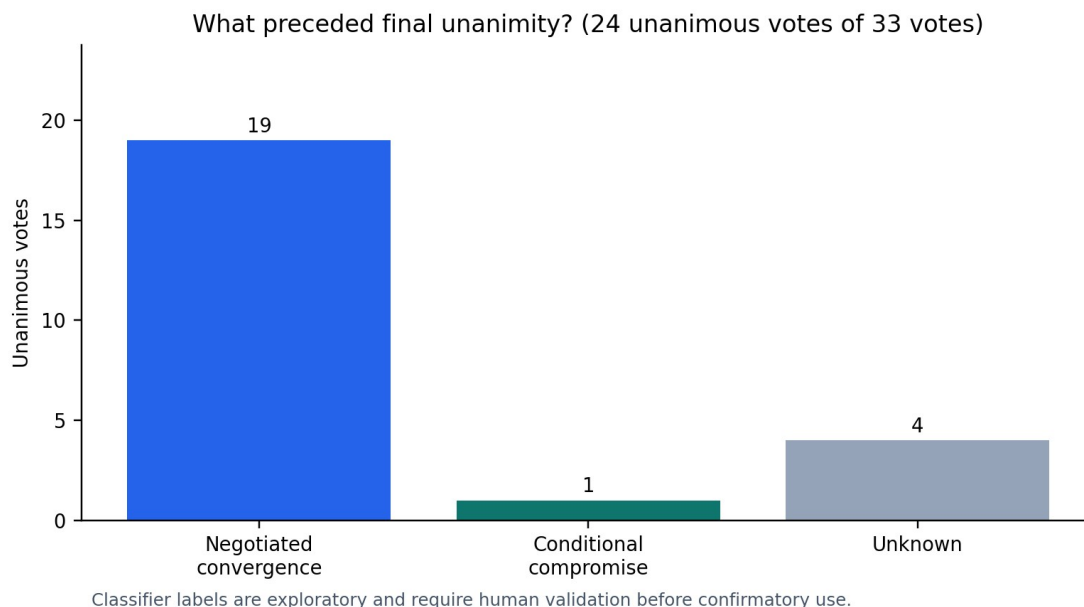


Figure 7. Current automated trajectory labels for the 24 unanimous final votes. Labels are exploratory pending human annotation and inter-rater validation.

The tiny-model pilot produced only 33 final votes because many governments lost members early. Of these, 24 were unanimous. The current automated trajectory classifier labels 19/24 as negotiated convergence, one as conditional compromise, and four as unknown. These counts show that the harness records trajectory information that final-vote labels discard, but they are not confirmatory evidence about negotiated convergence: the classifier is currently a measurement hypothesis and must be frozen, human-labeled, and evaluated for inter-rater agreement before use as a primary result.

3.7 Stated dispositions are not fixed policies

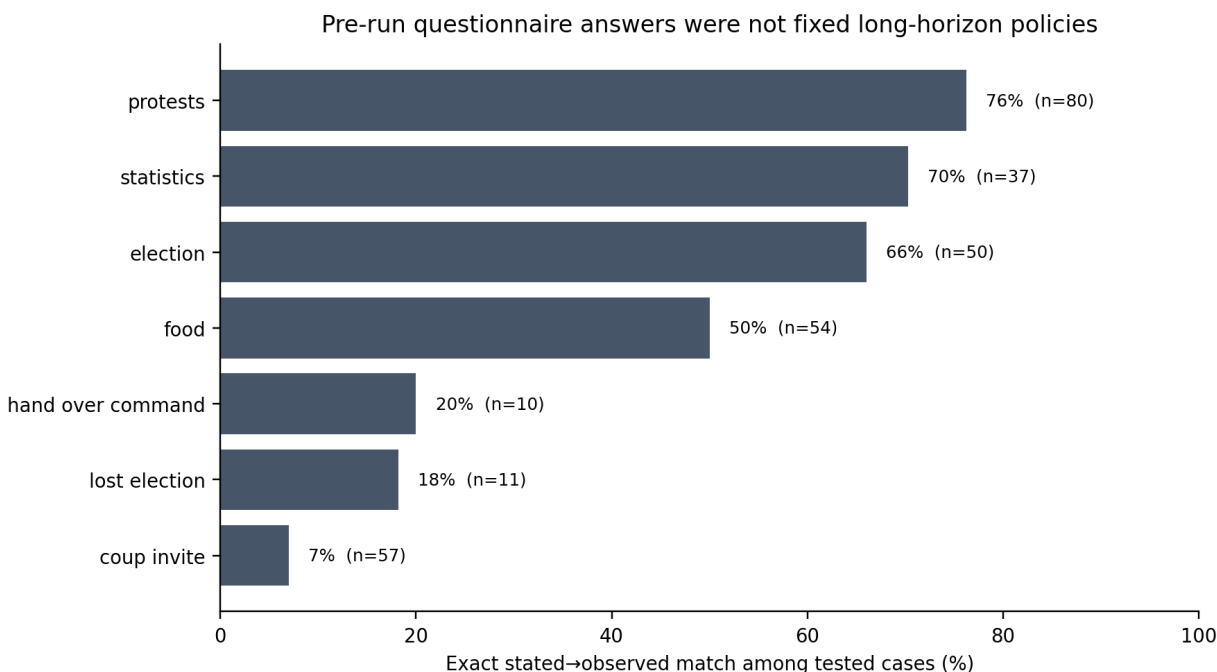


Figure 8. Exact match between pre-run questionnaire answer and later observed behavior for scenarios that actually occurred. Small denominators and evaluator validity require caution.

Before each run, delegates answered standardized hypothetical questions. Later behavior often diverged from those answers. Exact matches were relatively common for protest handling (61/80 tested cases) and publication of bad statistics (26/37), but rare when invited into a coup (4/57), ordered to hand over military command (2/10), or after losing an election (2/11). This is evidence that a pre-run questionnaire should be interpreted as an initial stated disposition, not as a ground-truth personality. The very low coup-invite match rate is sufficiently extreme that the evaluator mapping itself must be manually audited before substantive interpretation.

4. Harness validity and limitations

4.1 State survival currently overstates government functionality

The most important engine finding is that the fictional state can continue for long periods with most offices vacant. Several trajectories reached the scheduled election and constitutional handover while only one or two delegates remained active. At least one handover trajectory formed no functioning five-office cabinet at the beginning, yet the world still evolved to the election. Therefore, terminal survival or handover is not a valid proxy for governance competence in the current pilot.

Before confirmatory frontier runs, the engine should impose explicit vacancy consequences or a documented continuity mechanism: reduced implementation capacity, slower or blocked office-specific actions, administrative degradation, or a constitutional replacement process. The key requirement is not to make failure more frequent; it is to make office vacancy causally meaningful and auditable.

4.2 Behavioral integrity

Eighty-three of 85 completed reports are classified CLEAN by the harness. Two are CORRECTED_BEHAVIORAL, meaning a mismatch between natural-language intent and structured action reached canonical state. These two runs should remain in engineering archives but be excluded from clean behavioral inference unless deterministically replayed from the last clean checkpoint after the relevant validator is fixed.

4.3 Incomplete analytics

Promise follow-through is not yet usable as a headline metric. Across completed runs, the current promise evaluator labels 1,184 records AMBIGUOUS, 177 PENDING, 20 BROKEN, 18 EXPIRED, and only 17 KEPT. Bargain-followthrough records are effectively absent in this pilot. These fields should either be improved and validated or omitted from confirmatory claims.

4.4 Scope

The pilot evaluates language models inside a synthetic institutional environment. It does not establish that the models reproduce human political psychology, nor should results be used to predict elections, governments, or populations. More importantly, this first campaign should be read as an instrument-validation dataset: it demonstrates discriminative signal and surfaces failure modes, but several apparent behavioral effects may still be properties of the current action interface or engine semantics.

5. Relationship to prior work

Karamaniya inherits the broad idea of persistent generative agents but targets a different measurement problem. Generative Agents emphasizes believable memory-driven social behavior; Concordia emphasizes flexible generative agent-based modeling; SOTOPIA emphasizes social interaction and goal completion; GovSim isolates resource-governance cooperation; and AgentSociety emphasizes large-scale generative social simulation. Karamaniya is narrower in population size but deeper in institutional persistence: five agents repeatedly operate the same offices in a canonical world with private evidence, explicit voting, private bargaining, long-term memory, model-to-seat rotations, and an auditable action-to-execution pipeline.

The broader relevance is consistent with the emerging multi-agent safety agenda. In 2026, Google DeepMind and partners highlighted sandboxes/testbeds and the science of agent networks as priority areas for multi-agent safety research, including unexpected population-level properties and failure modes. Karamaniya is intended as one such reproducible testbed, with emphasis on institutional behavior rather than human-political forecasting.

6. Proposed Anthropic-supported study

Anthropic expanded its AI for Science program in August 2026 beyond the life sciences and describes support for ambitious, compute-intensive research, with credits of up to \$50,000 per project [1]. Eligibility for this application rests on the Official Rules [2], not on the announcement language. The Rules state that an individual applicant must be at least 18 years old or, if greater, the age of majority in the applicant's jurisdiction of residence at the time of application. They further state that institutional consent is required when an applicant is employed by or affiliated with an employer, university, or other research institution; such affiliation is not itself a universal eligibility prerequisite [2]. This proposal therefore applies as an independent individual researcher in Uzbekistan and would use the project organization/account structure required by the application and credit-provisioning process.

6.1 Preregistered hypotheses

- H1: within the Claude 5.5 family, time to institutional exit and cause-specific exit hazards differ between model classes and/or effort settings after office, seed, and scenario are controlled.
- H2: balanced seat swaps reveal model-by-office interactions that replicate across independent seeds.
- H3: final unanimous votes frequently follow measurable pre-vote disagreement or conditional bargaining.
- H4: structured-action reliability and institutional persistence are partially separable dimensions.
- H5: pre-run stated dispositions predict some low-stakes behaviors but weaken under high-stakes accumulated trajectories.

- H6: strengthening vacancy consequences and requiring structured resignation intent reduces false-positive institutional “survival/exit” signals without eliminating legitimate long-run variation.

6.2 Experimental phases

Phase	Design	Purpose
A. Instrument repair + bridge replay	Fix vacancy mechanics and resignation semantics; for ~25 matched tiny-model seeds, run paired bridge replays under the frozen corrected engine, including semantically equivalent action schemas with randomized/reversed field ordering; human-label a stratified sample.	Quantify measurement drift, test whether resign/exit rates are sensitive to action-menu order, and verify which persistence signals survive corrected instrumentation.
B. Frontier calibration + homogeneous Claude baselines	Run ~10 calibration trajectories first, then balanced homogeneous Claude Sonnet 5.5 / Opus 5.5 governments across cells and independent seeds.	Measure token growth, cache-hit ratio, model retention, and within-family scaling on the corrected engine.
C. Balanced seat-swap experiments	Place each target Claude model systematically in Head, Treasury, Interior, Army, and Navy against fixed references.	Separate model identity from institutional role with prespecified interaction tests.
D. Mechanism ablations	Remove or equalize memory, asymmetric information, relationships, and revision while holding the engine fixed.	Identify which architecture components cause divergence, convergence, or exit.
E. Preregistered holdout replication	Freeze hypotheses and analysis, then run untouched seeds/scenarios with human-validated trajectory labels.	Test whether exploratory effects replicate prospectively.

6.3 Primary endpoints

The primary persistence endpoint will be time to institutional exit, with exit cause recorded explicitly. Confirmatory analysis will use Kaplan-Meier curves for all-cause persistence, Aalen-Johansen cumulative-incidence estimates for competing outcomes (voluntary resignation, coup-related removal, revolution, constitutional handover, and other terminal events), and a prespecified cause-specific Cox model with robust run-level clustering; shared-frailty and proportional-hazards diagnostics will be reported as sensitivity analyses. Office, scenario, experimental cell, and model/effort setting will be prespecified covariates, with interaction terms limited to hypotheses supported by adequate cell counts. A Month-6 landmark may be reported secondarily for interpretability but will not define the main result. The second primary endpoint will

quantify disagreement hidden by final unanimity: $P(\text{initial material disagreement} \mid \text{final unanimous vote})$, using a frozen definition of material disagreement and blinded human validation. Secondary endpoints include protocol-error rate, vote-to-execution consistency, position changes, conditional support, unexplained reversals, and state degradation under office vacancies.

6.4 Reproducibility

Each research run will store exact model identifier and effort setting, prompt and architecture versions, world seed, model-call metadata, raw and parsed outputs, vote trajectories, validation findings, canonical state checkpoints, and the final report. Semantic engine changes will trigger a version bump and fresh prospective runs. Corrected behavioral runs will not be pooled with clean runs. Because Phase A changes vacancy and resignation semantics, pilot-v4 effect sizes will not be used as priors for confirmatory power calculations or directly pooled with frontier runs. Matched tiny-model bridge replays on the frozen engine will quantify measurement drift, and confirmatory hypotheses will be frozen before untouched holdout seeds are opened.

7. Compute justification and requested credits

The pilot provides a lower-bound compute trace, not a reliable frontier cost estimate. Across the 85 completed local runs, Karamaniya recorded 32.76 million input tokens and 4.22 million output tokens across 7,237 calls. The nine election-handover trajectories averaged 0.755 million input tokens, 0.100 million output tokens, and 159 calls per trajectory, but most had already lost several agents. Frontier governments that retain five agents for 18+ months will make more calls, carry larger memories, and may incur materially higher billed output at higher effort settings. The funded study will therefore begin with a ~10-trajectory Claude calibration batch. The confirmatory grid will be resized from observed full-government call counts, input/output tokens, cache utilization, and effort-setting overhead before the main campaign is locked.

For planning, the proposal assumes approximately 2-4x the observed long-run pilot token volume per full frontier trajectory before caching, then treats the calibration batch as authoritative. As of 1 October 2026, Claude Sonnet 5.5 is priced at \$2/MTok input and \$10/MTok output, while Claude Opus 5.5 is \$4/MTok input and \$20/MTok output; cache reads are \$0.20/MTok for both [3,4]. Anthropic also notes that higher effort settings generally increase cost per task because models reason for longer [3]. Karamaniya will cache immutable system/schema prefixes where technically appropriate, but no cache-hit or batch-discount rate is assumed in the planning ceiling. Sequential dependence remains essential within each trajectory: later prompts depend on earlier votes, messages, memories, office vacancies, and world consequences.

7.1 Transparent planning arithmetic

The table below makes the \$30,000 envelope auditable. The simple token-only range uses 2-4x the pilot handover average (0.755M input, 0.100M output per trajectory). The conservative ceiling uses the 4x input case and a 3x multiplier on the corresponding visible output-token volume to cover effort-driven billed-output growth. This is a planning allowance, not a claim about expected realized reasoning usage. Prompt caching and batch discounts, if achieved, only reduce spend from this ceiling.

Budget component	Scale	Per-trajectory planning assumption	Unit / reserve	Planning ceiling
Sonnet 5.5 trajectories	760	4x pilot long-run = 3.02M input + 0.40M visible output; ceiling allows 1.19M billed output at higher effort	\$18.0/run	\$13.7k
Opus 5.5 trajectories	150	Same conservative token envelope as Sonnet	\$36.0/run	\$5.4k

Budget component	Scale	Per-trajectory planning assumption	Unit / reserve	Planning ceiling
Clean reruns + evaluator redundancy	-	Integrity-triggered replay, duplicate human/evaluator passes, and failed-run replacement	25% of model-call ceiling	\$4.8k
Calibration / variance reserve	-	Protects against longer context growth or higher observed effort cost after the 10-run calibration	20% after rerun reserve	\$4.8k
Calculated ceiling	910 Claude trajectories	Before any prompt-cache or batch savings	-	~\$28.6k
Requested rounded envelope	-	Staged maximum; unused credits are not treated as required spend	-	\$30.0k

Arithmetic check. Without the higher-effort and reserve allowances, the same 910-trajectory split costs approximately \$5.3k at 2x pilot volume and \$10.6k at 4x pilot volume before caching. At the conservative ceiling above, 760 Sonnet runs contribute about \$13.7k and 150 Opus runs about \$5.4k; the remaining ~\$9.5k is an explicit rerun/evaluator plus calibration-variance reserve. The 10-run calibration is therefore a budget-control gate, not merely a pilot convenience.

Work package	Planned scale	Credit use
Frontier calibration + bridge checks	~10 Claude Sonnet 5.5 calibration trajectories; ~25 matched tiny-model replays on local compute	Measure full-government token growth, cache-hit ratio, and instrumentation shift before locking the confirmatory grid.
Homogeneous Sonnet 5.5 baselines	~300 trajectories	Core variance estimates and corrected-engine survival distributions.
Matched Opus 5.5 scale subset	~150 trajectories	Within-family model-class comparison on matched seeds/cells.
Seat swaps + mechanism ablations	~300 Sonnet 5.5 trajectories	Identify model x office effects and memory/information/revision mechanisms.
Holdout replication + evaluator/rerun reserve	~150 Sonnet 5.5 trajectories plus evaluator passes	Prospective confirmation, human-label redundancy, and clean reruns when integrity checks fail.

Requested compute envelope: \$30,000 in Claude API credits. The arithmetic below treats 760 trajectories as Sonnet 5.5 and 150 as Opus 5.5 (910 Claude trajectories total), with the ~25 bridge replays run locally. The base 2-4x token-only estimate is approximately \$5.3k-\$10.6k before caching. The \$30k request is a

conservative ceiling, not an expected spend: it protects against high-effort billed-output growth, integrity-triggered clean reruns, evaluator redundancy, and calibration variance. If the 10-run calibration predicts materially lower or higher cost, trajectory counts will be resized before confirmatory execution rather than exhausting credits mid-trajectory.

The research bottleneck is therefore replication depth and controlled frontier access, not simulator implementation. The local pilot has already demonstrated end-to-end execution, logging, analysis, and artifact generation; the requested credits would be spent on corrected, preregistered trajectories and measurement validation that cannot be approximated by isolated one-shot prompts.

8. Expected outputs and broader value

The proposed study is intended to produce a reproducible benchmark environment, a cleaned and versioned trajectory dataset, validated trajectory labels, and a research paper on persistent multi-agent behavior. The work is directly useful to evaluation of AI organizations and agent networks because it distinguishes model behavior from role effects and harness artifacts, and because it measures gaps between deliberation, final decisions, and downstream execution.

The most valuable outcome would not be a leaderboard. It would be a set of experimentally validated measurements for questions such as: when does a persistent agent abandon an institution, when does consensus reflect genuine convergence rather than procedural consolidation, how do role assignments change the behavior of the same model, and which apparent long-horizon behaviors disappear when memory or information asymmetry is removed?

9. Conclusion

The local pilot turns Karamaniya from a concept into a testable measurement instrument. Its strongest result is not that one tiny model “governs better” or is intrinsically more persistent than another; it is that the harness produces large, repeatable between-model differences while simultaneously exposing the interface and engine confounds that could create false behavioral interpretations. Observed time-to-exit differs sharply across models, candidate role effects are visible, and structured-protocol reliability is not identical to observed persistence - but resignation semantics, action-order sensitivity, and vacancy coasting must be repaired before those signals can support a benchmark claim. The next stage is therefore a bridge-and-validation study: freeze the corrected instrument, test resignation intent and action-order robustness, verify vacancy consequences, validate trajectory labels, preregister competing-risk survival endpoints, calibrate Claude compute, and replicate on untouched seeds. Anthropic compute credits would make that transition from exploratory instrument validation to controlled frontier evidence possible.

Appendix A. Exploratory survival-model robustness summary

Model term (reference: DeepSeek-R1 1.5B)	Hazard ratio	95% CI	Exploratory p
Qwen3.5 0.8B	0.82	0.63-1.08	0.162
SmolLM2 1.7B	2.28	1.68-3.10	1.14e-7
Gemma3 1B	4.53	3.31-6.19	2.61e-21
Llama3.2 1B	4.80	3.56-6.48	9.34e-25

The table below reports the model terms from an exploratory Cox proportional-hazards model of all-cause institutional exit in the 75-run strict core. DeepSeek-R1 1.5B is the model reference category; the regression also includes founding office and experimental cell, and standard errors are clustered by run. These estimates were not preregistered, proportional-hazards assumptions have not been used for confirmatory inference, and p-values are uncorrected for multiplicity. The planned study replaces this post-hoc analysis with a preregistered competing-risk survival model.

Appendix B. Data-quality notes

The 85 completed runs contain 83 CLEAN behavioral-integrity reports and two CORRECTED_BEHAVIORAL reports. Complete five-office formation occurred in 77/85 runs; the overlap of CLEAN and complete formation is 75 runs. All main persistence figures use that 75-run strict core. The broader 85-run set is used for terminal-outcome and engineering diagnostics where appropriate.

Automated promise metrics are excluded from substantive conclusions because most records are AMBIGUOUS or PENDING. Automated unanimity classes and pre-run-versus-observed behavior mappings are presented as exploratory and require human validation before confirmatory publication.

Resignation is a direct boolean action in pilot-v4. The post-hoc support-text audit is intentionally weak and should not be interpreted as proof of intent. The pilot did not randomize action/field order. The corrected engine will require explicit resignation rationale/trigger fields, two-step confirmation, and a paired field-order randomization diagnostic on matched seeds before exit-rate differences are treated as behavioral. One completed run also contains malformed raw JSONL fragments in a foreign-call record; report/checkpoint aggregates remain readable, but that trace is excluded from raw-line analyses.

References

- [1] Anthropic. (2026). Expanding our support for scientists. 27 August 2026. <https://www.anthropic.com/news/expanding-support-for-scientists>
- [2] Anthropic. (2026). AI for Science Program Official Rules. Updated 25 August 2026. <https://www.anthropic.com/ai-for-science-program-rules>
- [3] Anthropic. (2026). Claude Sonnet 5.5. 28 September 2026. <https://www.anthropic.com/claude-sonnet-5-5>
- [4] Anthropic. (2026). Claude Opus 5.5. 22 September 2026. <https://www.anthropic.com/claude-opus-5-5>
- [5] Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative Agents: Interactive Simulacra of Human Behavior. arXiv:2304.03442.
- [6] Vezhnevets, A. S., et al. (2023). Generative agent-based modeling with actions grounded in physical, social, or digital space using Concordia. arXiv:2312.03664.
- [7] Zhou, X., et al. (2023). SOTOPIA: Interactive Evaluation for Social Intelligence in Language Agents. arXiv:2310.11667.
- [8] Piatti, G., Jin, Z., Kleiman-Weiner, M., Schölkopf, B., Sachan, M., & Mihalcea, R. (2024). Cooperate or Collapse: Emergence of Sustainable Cooperation in a Society of LLM Agents. NeurIPS 2024.
- [9] Piao, J., et al. (2025). AgentSociety: Large-Scale Simulation of LLM-Driven Generative Agents Advances Understanding of Human Behaviors and Society. arXiv:2502.08691.
- [10] Google DeepMind, Schmidt Sciences, Cooperative AI Foundation, ARIA, & Google.org. (2026). Investing in multi-agent AI safety research. 11 June 2026. <https://deepmind.google/blog/investing-in-multi-agent-ai-safety-research/>